

Derbyshire Shared Care Pathology Guidelines

MGUS (Monoclonal Gammopathy of Undetermined Significance)

Purpose of guideline

This guideline provides information about the risk stratification of patients diagnosed with MGUS and their management.

This guideline should be read in conjunction with shared care pathology guidelines for myeloma screening in primary care.

Definition

The presence of a monoclonal protein (also known as a paraprotein), in the serum or urine of an individual with no evidence of multiple myeloma, AL amyloidosis, Waldenstrom macroglobulinaemia or other related disorder.

Paraproteins can be detected in the serum of about 1% of the population, and are frequently detected as a result of a myeloma screen (see shared care guideline for myeloma for further information). MGUS is uncommon below the age of 50 years and the prevalence increases with age. It is important to identify and monitor those patients at highest risk of progression to significant disease, whilst avoiding unnecessary follow-up of patients at low risk of progression.

Diagnostic criteria for MGUS

- Paraprotein in serum <30g/L
- Bone marrow clonal plasma cells <10%
- No myeloma-related organ or tissue impairment (CRAB criteria-see below)
- No evidence of CLL or lymphoma,
- No suggestion of AL amyloidosis (unexplained heavy proteinuria, heart failure with preserved ejection fraction peripheral neuropathy or bilateral carpal tunnel syndrome)

Risk stratification for MGUS

The risk of progression of MGUS into multiple myeloma or lymphoma is small (around 1% per year) and therefore most patients will not progress. See appendix 1 for details on risk of progression.

The lab will provide a guidance on the interpretation of the protein electrophoresis and the light chains results including advice on referral. It is important to view this interpretation in clinical context and the ultimate responsibility of making a referral to the haematologist lies with the primary care physician.

Referral criteria

Paraprotein present

+

Any **Unexplained**:

- Hypercalcaemia
- Renal Impairment and/or heavy proteinuria
- Anaemia
- Lytic lesions/Osteoporotic fractures
- Weight loss
- Bilateral Carpal Tunnel syndrome
- Symptoms of hyperviscosity

And/or

IgG/IgA Paraprotein >30 g/L (+/-)

IgM Paraprotein > 15 g/L

Very Abnormal FLC ratio (>8 or <0.125)

Yes

**Urgent 2WW Consultant
Haematology Referral. ***

No

IgD or IgE paraprotein

Any level

IgA or IgM paraprotein

≥ 10 g/L

IgA or IgM paraprotein

< 10g/L

Light chain ratio ≤0.25 or ≥4

IgG paraprotein

≥ 15 g/L

IgG paraprotein

< 15 g/L

Light chain ratio ≤0.25 or ≥4

**Routine referral
to Haematology**

No

**Community monitoring **
Or
Routine referral to CRH
Or
Seek advice if unclear**

Please note

- The level of the paraprotein is a distinct measurement from the total immunoglobulins level. For referral purposes, the level of the paraprotein should be used rather than the total immunoglobulin level.
- The presence of an **IgM** paraprotein may be associated with lymphoma or other lymphoproliferative disorders – please check for B symptoms including weight loss, fever, night sweats, lymphadenopathy or organomegaly.
- Any patient with symptoms, signs or results suggestive of myeloma, other lymphoproliferative disorders or AL amyloidosis needs to be dealt with outside of this stratification listing, and needs to be seen by a Consultant Haematologist.

Urgent Haematology referral group

Make an URGENT 2WW referral for these patients to be seen by a Consultant Haematologist for further assessment

*If new/significant renal impairment consider renal referral in addition.

Routine Haematology referral group

Make a routine referral for these patients to be seen by a Consultant Haematologist for further assessment. They will then be followed up either in the Consultant or CNS led clinic.

****Community Monitoring group.**

These patients have the lowest transformation rate, and can be managed well in the community as detailed below, without the need for further invasive testing or Consultant referral. There are currently 2 pathways, one for South Derbyshire and one for North Derbyshire:

(i) South Derbyshire / referral to UHDB

Follow-up as detailed below. If in doubt, consider advice and guidance referral or seeking phone advice.

(ii) North Derbyshire / referral to CRH

The Haematology team at Chesterfield Royal Hospital have to date taken these referrals, assessed the patient and then discharged back to Primary Care for monitoring. However, if you are happy to follow the process already established for South Derbyshire, please do so. If you need advice, please use the CRH email advice service at crhft.cabadviceandguidance@nhs.net Otherwise, please make a routine referral to the Haematology clinic at CRH.

Follow-up for patients monitored in Community:

Patients should be assessed every 6 months initially by being asked specifically about bone pain and episodes of infection, then annually if stable. Patients do not need routine examination unless there is a change in symptoms.

Blood tests should be carried out prior to each assessment as follows:

- Protein electrophoresis
- Serum free light chains
- Full blood count
- Urea and electrolytes
- Corrected calcium
- Immuglobulins

These can be selected using the MGUS monitoring button on ICE

Patients should be re-referred to the consultant clinic as below:

- New Unexplained symptoms weight loss, night sweats, persistent fever, lymphadenopathy or organomegaly
- Bone symptoms New bone pain or compression fracture without other cause
- Infection >2 significant bacterial infections within 12 months (excluding UTI) *e.g. sepsis or proven bacterial LRTI (with no underlying lung pathology) requiring hospital admission.*
- Anaemia Hb <100g/L or 20g/L below reference range with no blood loss, normal haematinics
- Renal Function >25% increase in creatinine **OR** creatinine >173 umol/L without other cause
- Hypercalcaemia Increase in calcium to >2.75 mmol/L (with suppressed PTH)
- Paraprotein >25% increased concentration **AND** total increase ≥ 5g
- FLC ratio Development of Kappa:Lambda ratio of ≥4.0 or ≤0.25

Patient information

An excellent patient information sheet is available on the Myeloma UK website:
<https://www.myeloma.org.uk/wp-content/uploads/2018/03/Myeloma-UK-MGUS-Infosheet.pdf>

References

1. UK Myeloma Forum (UKMF) and Nordic Myeloma Study Group (NMSG): guidelines for the investigation of newly detected M-proteins and the management of monoclonal gammopathy of undetermined significance (MGUS). British Journal of Haematology 2009, **147**, 22-42.
2. Eythorsson E et al. Development of a Multivariable Model to Predict the Need for Bone Marrow Sampling in Persons With Monoclonal Gammopathy of Undetermined Significance Ann Intern Med. 2024;177:449-457. doi:10.7326/M23-2540
3. Rajkumar SV et al. Serum free light chain ratio is an independent risk factor for progression in monoclonal gammopathy of undetermined significance. Blood 2005, 106 (3), 812-817.

Useful Contacts

UHDB Consultant Haematologist	01332 787973 (secretary)
UHDB Clinical Nurse Specialist	07825 342918
CRH Consultant Haematologist	crhft.cabadviceandguidance@nhs.net#
QHB Consultant Haematologist	01283 511511 ext. 4025/4032 (Secretary)
UHDB Duty Biochemist	01332 789383
CRH Duty Biochemist	01246 512212

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	V1 July 2011	Dr David Allotey, Dr Chris Millar, Dr Nigel Lawson	New Document
	V2 Oct 2013	Dr D Allotey, Dr P Blackwell, Mrs H Seddon	Routine Review
	V3 Oct 2015	Dr D Allotey, Dr S Hebballi, Dr I Amott, Mrs H Seddon	Routine review
	V4 Jan 2018	Dr D Allotey, Dr P Blackwell, Mrs H Seddon	Routine review
	V5 Apr 2020	Dr R Faulkner, Dr I Amott, Dr P Blackwell, Ms J Forsyth, Mrs H Seddon	Routine review
	V6 Sep 2022	Dr R Faulkner, Dr I Amott, Dr P Blackwell, Mrs H Seddon	Routine review
	V7 Dec 2024	Dr R Faulkner, D F Al-Kaisi, Dr J Tam, Dr K Lam, Dr P Blackwell, Ms S Knowles	Routine review
Approval	Date	Name(s)	Designation
Secondary Care UHDB	14.08.25	Dr F Al Kaisi	Consultant Haematologist
Secondary Care CRH	11.08.25 14.08.25	Dr K Lam, Dr J Tam	Consultant Haematologists
Primary Care	18.08.25	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	18.08.25	Ms S Knowles	Consultant Clinical Scientist

Appendix 1 Risk stratification for MGUS

Score one for each of the points below and calculate the total number of points

- Paraprotein that is NOT of an IgG subtype **(1 point)**
- Paraprotein level > 15 (regardless of the subtype) **(1 point)**
- Abnormal Serum free light chain ratio **(1 point)**

Total points	Risk category	Absolute risk of progression at 20 years (%)
0 points	Low risk	5%
1 point	Low-intermediate risk	21%
2 points	High-intermediate risk	37%
3 points	High risk	58%

Reference: [Advances in the Diagnosis, Classification, Risk Stratification, and Management of Monoclonal Gammopathy of Undetermined Significance: Implications for Recategorizing Disease Entities in the Presence of Evolving Scientific Evidence - PMC](#)